WORKED SOLUTION

STEP III 1996, Question 6

Pure • Groups

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 6.8, *A Group of Singular Matrices*: https://beyondstep.keithmcnulty.org/ch06/singular-group.html

The key idea

Every matrix in S is a multiple of one fixed matrix $\mathbf{J}$, and $\mathbf{J}^2$ is again a multiple of $\mathbf{J}$, so multiplying elements of S comes down to multiplying real numbers. The surprise is that the identity of this group is *not* the identity matrix $\mathbf{I}$: the group never needs $\mathbf{I}$ at all. Part (ii) explains why that has to happen: inside a group, every element can be written as a product involving any other, and determinants multiply.

Solution

Part (i)

Write

$$\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \text{so that} \qquad \mathbf{J}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2\mathbf{J}.$$

Every element of S is $a\mathbf{J}$ with $a \neq 0$, and we write $\mathbf{M}_a = a\mathbf{J}$.

Closure. For non-zero a and b,

$$\mathbf{M}_a\mathbf{M}_b = ab\,\mathbf{J}^2 = 2ab\,\mathbf{J} = \mathbf{M}_{2ab}.$$

Since $a \neq 0$ and $b \neq 0$, we have $2ab \neq 0$, so the product lies in S. Hence S is closed under matrix multiplication.

Now we check the remaining group axioms.

Associativity. Matrix multiplication is associative for all 2×2 matrices, so in particular for those in S.

Identity. We want $\mathbf{M}_e$ with $\mathbf{M}_e\mathbf{M}_b = \mathbf{M}_b$ for every b, that is, $2eb = b$. So we take $e = \frac{1}{2}$:

$$\mathbf{M}_{1/2}\mathbf{M}_b = \mathbf{M}_{2\cdot\frac{1}{2}b} = \mathbf{M}_b,$$

and similarly $\mathbf{M}_b\mathbf{M}_{1/2} = \mathbf{M}_b$. So $\mathbf{E} = \frac{1}{2}\mathbf{J}$, which lies in S, is an identity element.

> **Ask yourself** The identity matrix $\mathbf{I}$ isn't in S. Doesn't a group of matrices need it?

No. A group needs *some* element that acts as an identity on the other elements of the set, and $\frac{1}{2}\mathbf{J}$ does that job for every element of S. It isn't an identity for all 2×2 matrices (for example $\frac{1}{2}\mathbf{JI} \neq \mathbf{I}$), but $\mathbf{I}$ isn't in S, so that doesn't matter.

Inverses. Given $\mathbf{M}_a$, we want $\mathbf{M}_c$ with $2ac = \frac{1}{2}$, so $c = \dfrac{1}{4a}$, which exists and is non-zero because $a \neq 0$. Then

$$\mathbf{M}_a\mathbf{M}_{1/(4a)} = \mathbf{M}_{1/(4a)}\mathbf{M}_a = \mathbf{M}_{2a/(4a)} = \mathbf{M}_{1/2} = \mathbf{E}.$$

All four axioms hold, so S is a group under matrix multiplication. (It is even abelian, since $\mathbf{M}_a\mathbf{M}_b = \mathbf{M}_{2ab} = \mathbf{M}_b\mathbf{M}_a$.) Note that every element of S has determinant $a^2 - a^2 = 0$. These matrices have no inverse in the ordinary sense, yet each has an inverse within the group.

Part (ii)

Suppose $\mathbf{A} \in G$ has $\det\mathbf{A} = 0$, and let $\mathbf{B}$ be any element of G. Because G is a group, $\mathbf{A}$ has an inverse $\mathbf{A}'$ in G with $\mathbf{AA}' = \mathbf{A}'\mathbf{A} = \mathbf{E}$. (We write $\mathbf{A}'$, not $\mathbf{A}^{-1}$: it is the inverse inside G, not the matrix inverse, which doesn't exist.)

Let $\mathbf{C} = \mathbf{A}'\mathbf{B}$. By closure, $\mathbf{C} \in G$, and

$$\mathbf{AC} = \mathbf{A}(\mathbf{A}'\mathbf{B}) = (\mathbf{AA}')\mathbf{B} = \mathbf{EB} = \mathbf{B},$$

using associativity and the fact that $\mathbf{E}$ is the identity of G. So the equation $\mathbf{AC} = \mathbf{B}$ has a solution $\mathbf{C}$ in G. Taking determinants, and using $\det(\mathbf{XY}) = \det\mathbf{X}\det\mathbf{Y}$,

$$\det\mathbf{B} = \det\mathbf{A}\ \det\mathbf{C} = 0 \cdot \det\mathbf{C} = 0.$$

Since $\mathbf{B}$ was arbitrary, every element of G is singular. In particular the identity $\mathbf{E}$ of such a group is singular, so it can't be $\mathbf{I}$, just as we saw in part (i).

> **Ask yourself** Where did we use that G is a group, rather than just a set closed under multiplication?

We used the inverse $\mathbf{A}'$ and the identity $\mathbf{E}$ to write $\mathbf{B}$ as $\mathbf{A}$ times something in G. Without them the result fails. For example, $\{\mathbf{I}, \mathbf{0}\}$ is closed under multiplication and contains a singular matrix, but $\mathbf{I}$ is not singular. (It isn't a group: $\mathbf{0}$ has no inverse.)

An example with $n = 3$. We copy part (i). Let

$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad G = \{\, a\mathbf{K} : a \in \mathbb{R},\ a \neq 0 \,\}.$$

Each entry of $\mathbf{K}^2$ is $1+1+1 = 3$, so $\mathbf{K}^2 = 3\mathbf{K}$ and

$$(a\mathbf{K})(b\mathbf{K}) = 3ab\,\mathbf{K}.$$

- *Closure:* if $a, b \neq 0$ then $3ab \neq 0$, so the product is in G.
- *Associativity:* this is inherited from matrix multiplication.
- *Identity:* $\mathbf{E} = \frac{1}{3}\mathbf{K}$, since $(\frac{1}{3}\mathbf{K})(b\mathbf{K}) = (b\mathbf{K})(\frac{1}{3}\mathbf{K}) = 3 \cdot \frac{1}{3}b\,\mathbf{K} = b\mathbf{K}$.
- *Inverses:* the inverse of $a\mathbf{K}$ is $\dfrac{1}{9a}\mathbf{K}$, since $3 \cdot a \cdot \dfrac{1}{9a} = \frac{1}{3}$.

So G is a group under matrix multiplication. Every $a\mathbf{K}$ has three equal rows, so $\det(a\mathbf{K}) = 0$: this is a group of singular 3×3 matrices.

(Another example that works just as well is the set of matrices $\mathrm{diag}(a, b, 0)$ with $a, b \neq 0$, whose identity is $\mathrm{diag}(1, 1, 0)$.)

Checks

- Test a case: $\mathbf{M}_3\mathbf{M}_{1/12}$ has every entry $3 \cdot \frac{1}{12} + 3 \cdot \frac{1}{12} = \frac{1}{2}$, so it equals $\mathbf{E} = \mathbf{M}_{1/2}$, as the inverse formula $\frac{1}{4a}$ predicts.
- We checked with exact fractions, for hundreds of random non-zero a, b, c, that the closure, identity and inverse formulas hold for both the 2×2 and the 3×3 families, that all these matrices have determinant 0, and that $\mathbf{A}(\mathbf{A}'\mathbf{B}) = \mathbf{B}$.

Common slips

- Taking $\mathbf{I}$ as the identity, or $\mathbf{A}^{-1}$ as the ordinary matrix inverse. Neither exists in S. Find the identity and inverses from the group's own multiplication rule.
- Forgetting to check that $2ab \neq 0$. Closure needs the product to have the right form *and* a non-zero parameter.

Beyond STEP

A group is defined by its operation, not by where its elements happen to live. Here a group sits inside the $n \times n$ matrices without sharing their identity. In fact $\mathbf{E}$ always satisfies $\mathbf{E}^2 = \mathbf{E}$, so it is a projection, and the group acts as invertible maps on the image of $\mathbf{E}$. The map $a \mapsto \mathbf{M}_{a/2}$ is an isomorphism from the non-zero reals under

multiplication to S, which you can check directly. For more, see Allenby, *Rings, Fields and Groups*, and for projections, Strang, *Introduction to Linear Algebra*.

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